Industrial Equipment Anomaly & Incident Response Agent

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Overview

We developed an agentic workflow (Plaat et al., 2025; Hughes et al., 2025) to classify pre-detected anomalies in industrial telemetry into three distinct operational response classes. The core system goal is to automate incident triage by identifying failure modes, conducting root-cause analysis, recommending corrective actions, and routing incidents to the appropriate operational tier for execution.

Built using LangGraph (LangChain Inc., 2026), the orchestration framework utilizes a single central Large Language Model (LLM) (Wikipedia contributors, 2026b) acting as a ReAct (Reasoning and Acting) agent (Yao et al., 2023) within the classifier node to manage workflow decisions.

Monitored Telemetry Parameters

The system evaluates four core telemetry parameters representative of critical industrial assets, such as a Multistage Centrifugal Water Pump (Wikipedia contributors, 2026a):

1. **Temperature (`temperature_c`):** Monitored at bearings, lubrication loops, or process fluids to track thermal friction and cooling efficiency.

2. **Vibration (`vibration_hz`):** Tracks mechanical oscillations to detect shaft misalignment, rotor imbalance, or bearing degradation.

3. **Pressure (`pressure_psi`):** Measures internal suction, discharge, or hydraulic line pressures.

4. **Power (`power_kw`):** Tracks real-time electrical power consumption to evaluate motor load and overall efficiency.

System Architecture and Design

Workflow Graph

The agentic system's execution flow is modeled as a directed orchestration graph shown in Fig. 1.

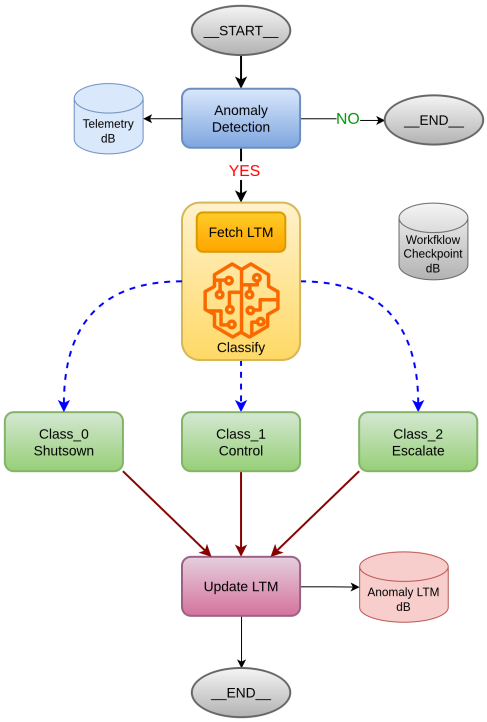


Figure 1

Workflow Graph.

The execution sequence follows a multi-stage routing pipeline:

1. **Anomaly Detector Node:** Telemetry data enters the anomaly detector node and is logged to the telemetry database. If no anomaly is detected, the workflow terminates. If an anomaly is present, data is forwarded to the Classifier Agent.

2. **Classifier Agent Node:** Operates as a ReAct agent using few-shot prompting (Gadesha, 2026) with clearly defined response classes. Context is enriched using the 3 most recent anomaly records retrieved from Long-Term Memory

(LTM) (Stryker, 2026; Chugani, 2025). The agent utilizes a single tool to fetch standardized severity bands for the current asset.

3. **`CLASS_0` Node:** Executes immediate asset shutdown protocols for critical safety hazards.

4. **`CLASS_1` Node:** Triggers automated control adjustments (e.g., via Variable Frequency Drive(VFD) (Wikipedia contributors, 2026d)) for controllable process drifts.

5. **`CLASS_2` Node:** Escalate complex or unhandled anomalies by generating a comprehensive report for an on-site engineer.

6. **LTM Update Node:** Updates the Long-Term Memory database with event details across all anomaly paths.

Data Storage and Persistence Architecture

To maintain zero external deployment overhead while preserving operational isolation, the system employs three light-weight SQLite databases:

- **Telemetry Database:** Stores continuous time-series logging of all incoming asset telemetry.
- **Short-Term Memory (STM) Checkpoint Store:** Captures full state checkpoints for offline auditing, system analytics, and performance tracking.
- **Long-Term Memory (LTM) Store:** Maintains a historical record of past anomaly events and resolutions per asset to provide historical context to the agent.
- **System Logging:** The system logs (error, debug, warning, info) are saved in **JSONL** format for easy offline parsing, and troubleshooting.

Please see sample of log and databases in the notebook.

Simplified Domain Grounding via Severity Bands

While a production system could integrate Retrieval-Augmented Generation (RAG) (Wikipedia contributors, 2026c) against technical manuals, this implementation uses a deterministic threshold tool to convert raw numerical anomalies into standardized severity bands (*Critically Low, Low, Normal, High, Critically High*), as shown in Fig. 2.

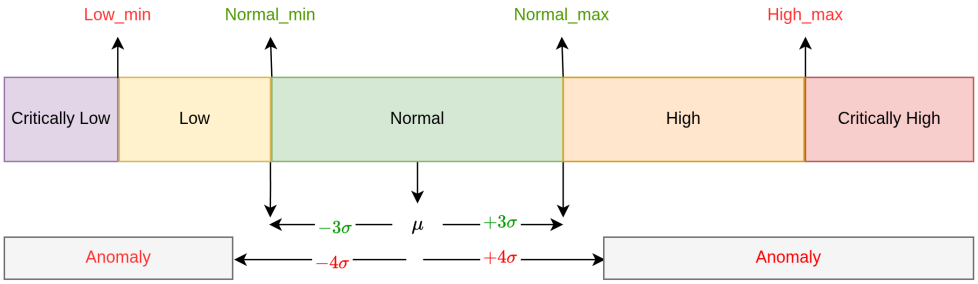


Figure 2

Severity Bands.

Bypassing raw statistical metrics (e.g., standard deviations and means) inside the LLM prompt prevents hallucination and reasoning errors. This abstraction aligns with practical engineering realities where a deviation of 10σ might represent a tolerable operational shift in one parameter but an immediate catastrophic failure in another.

Decision Logic and Behavior

Anomaly Detection Logic

Rather than relying on complex machine learning models, the anomaly detection node utilizes a transparent statistical rule set defined around baseline as follow (not necessarily a perfect Gaussian distributions):

- The full normal operating range spans $\pm 3\sigma$ centered around the mean.
- An anomaly is flagged whenever any metric exceeds $\pm 4\sigma$.
- The ranges $[-4\sigma, -3\sigma]$ and $[+3\sigma, +4\sigma]$ serve as operational tolerance buffers prior to reporting an anomaly.

Incident Severity Classification

Once an anomaly is flagged, the central classifier agent categorizes the incident into one of three operational classes:

0. `CLASS_0` **(Immediate Shutdown):** Triggered by severe threshold breaches (Critically High) on any dangerous parameter (temperature, pressure, or vibration) requiring instant asset shutdown to prevent catastrophic damage.

1. `CLASS_1` **(Automated Control):** Triggered by minor single-parameter operational stress or controllable process drift solvable via automated actuators (e.g. VFD).

2. `CLASS_2` **(Escalate to Onsite Engineer):** Triggered by physical degradation or complex multi-telemetry anomalies requiring physical inspection, maintenance, or manual intervention (all cases not fall in CLASS_0 or CLASS_1).

Safety, Reliability, and Transparency

Context Window Control and Latency Optimization

Response latency is critical in industrial automation. To minimize inference delay, LTM access is kept outside the agent's tool execution loop; instead, relevant LTM context (the last 3 historical anomalies) is directly pre-loaded into the prompt before invocation.

Furthermore, because state checkpoints in STM are reserved strictly for offline auditing and analytics, input context lengths remain deterministic. Complex context management techniques (Carrascosa, 2026)—such as sliding-window truncation or message summarization—are unnecessary, preventing context drop and ensuring consistent reasoning across runs.

System Observability and Analytics

The graph checkpointing mechanism captures system state after every node transition, enabling thorough post-hoc auditing. Figures 3 illustrate the offline tracking metrics extracted from checkpoint store, covering LLM tool call frequencies, cumulative token consumption, and node execution profiles (python function usage).

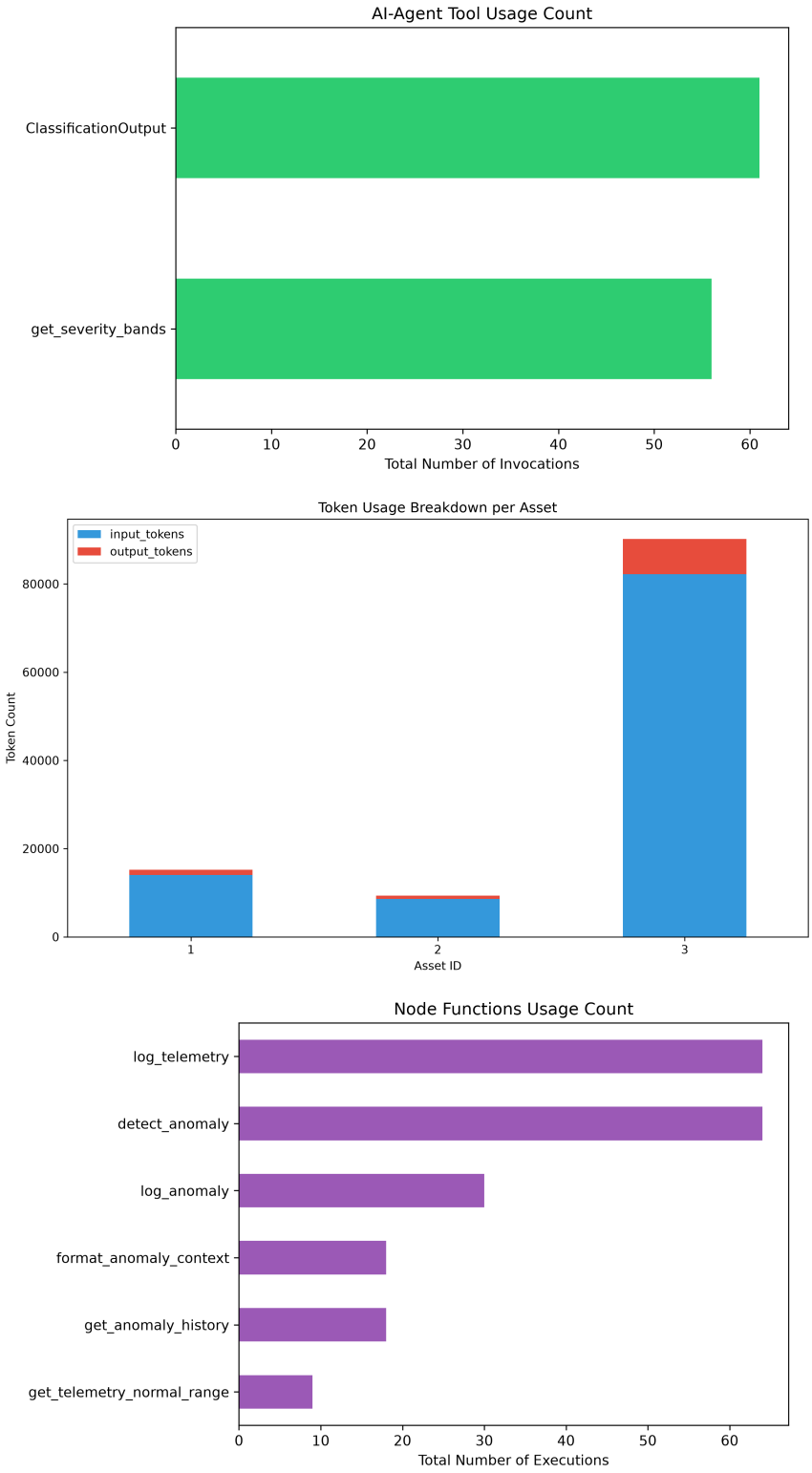


Figure 3

System Analytics: tool, token, and function usage.

Observed Behavior and Limitations

Experimental Evaluation and Sample Results

The framework was evaluated across 3 simulated centrifugal pump assets across 9 total test scenarios (see notebook for all results, and details in Appendix A: Asset and Test-case Info). Telemetry trends were simulated using sigmoid-like curve transitions for realistic parameter drift. Across baseline runs, the system executed expected actions across all classes, as illustrated in Fig. 4 (`CLASS_0`), Fig. 5 (`CLASS_1`), and Fig. 6 (`CLASS_2`).

LTM-Driven Behavioral Adaptation and Inconsistencies

To assess the impact of Long-Term Memory, a `CLASS_1` candidate scenario was evaluated under different LTM states:

- **Without LTM:** The agent consistently routed the event to `CLASS_1` for automated adjustment (Fig. 7).
- **With LTM (Identical Past Failures Loaded):** Upon observing in LTM that prior automated adjustments had failed to clear the anomaly, the agent adapted its behavior and escalated the event to `CLASS_2` (Fig. 8).
- **Repeated Run After Time Delay:** In subsequent runs, the agent attempted an automated control intervention (`CLASS_1`) once before re-escalating to `CLASS_2` on the next occurrence (Fig. 9).

While this shift demonstrates adaptive decision-making (giving automated control a secondary trial before escalating), it also highlights potential non-deterministic reasoning in LLM-based controllers.

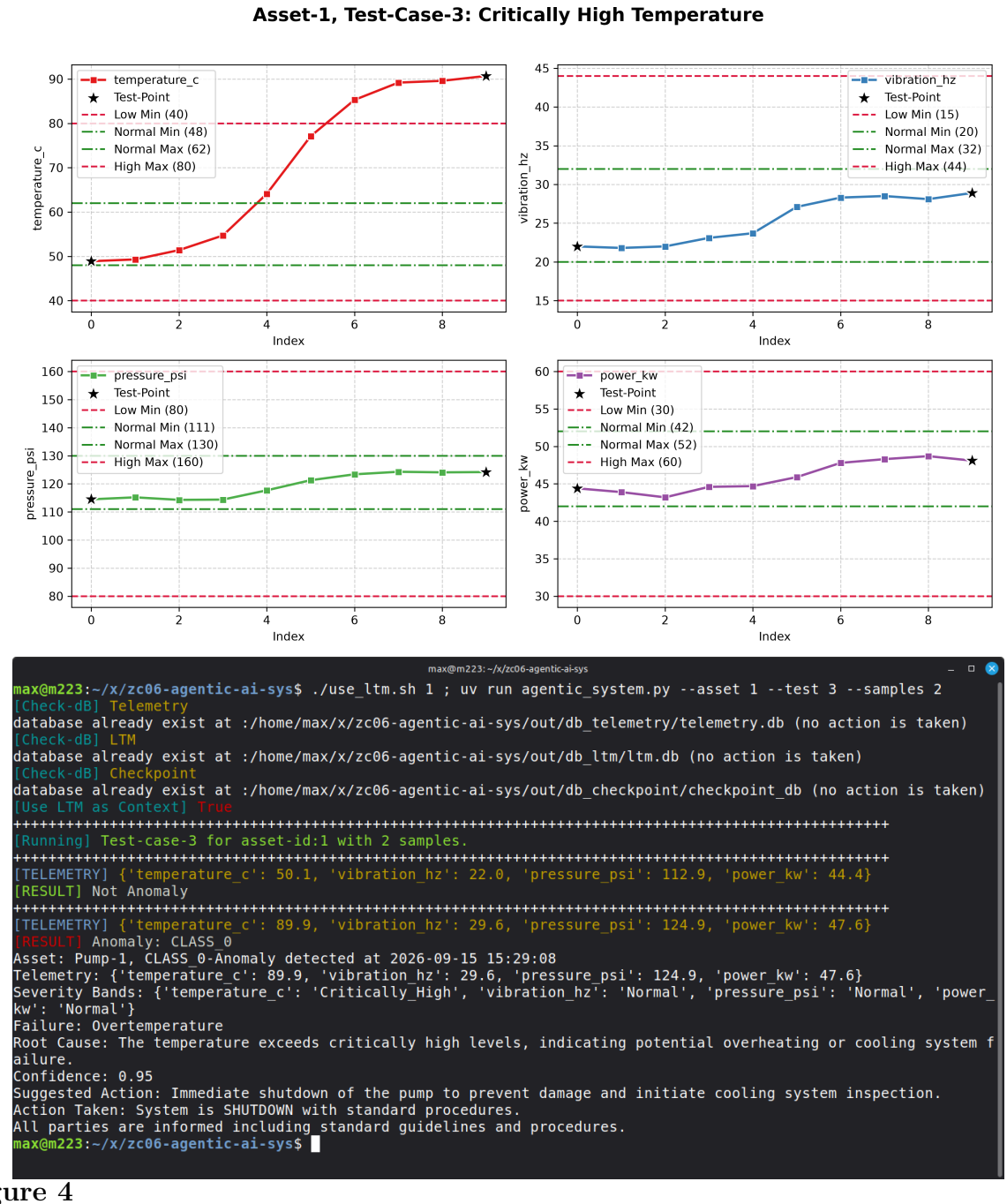
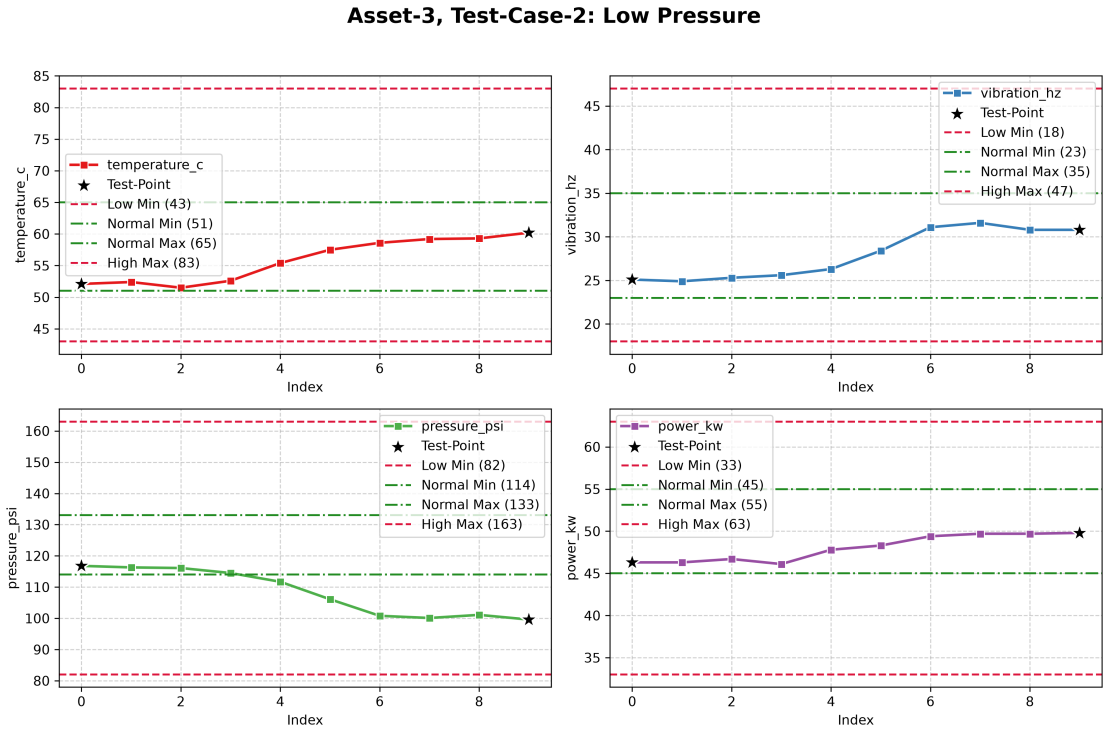


Figure 4

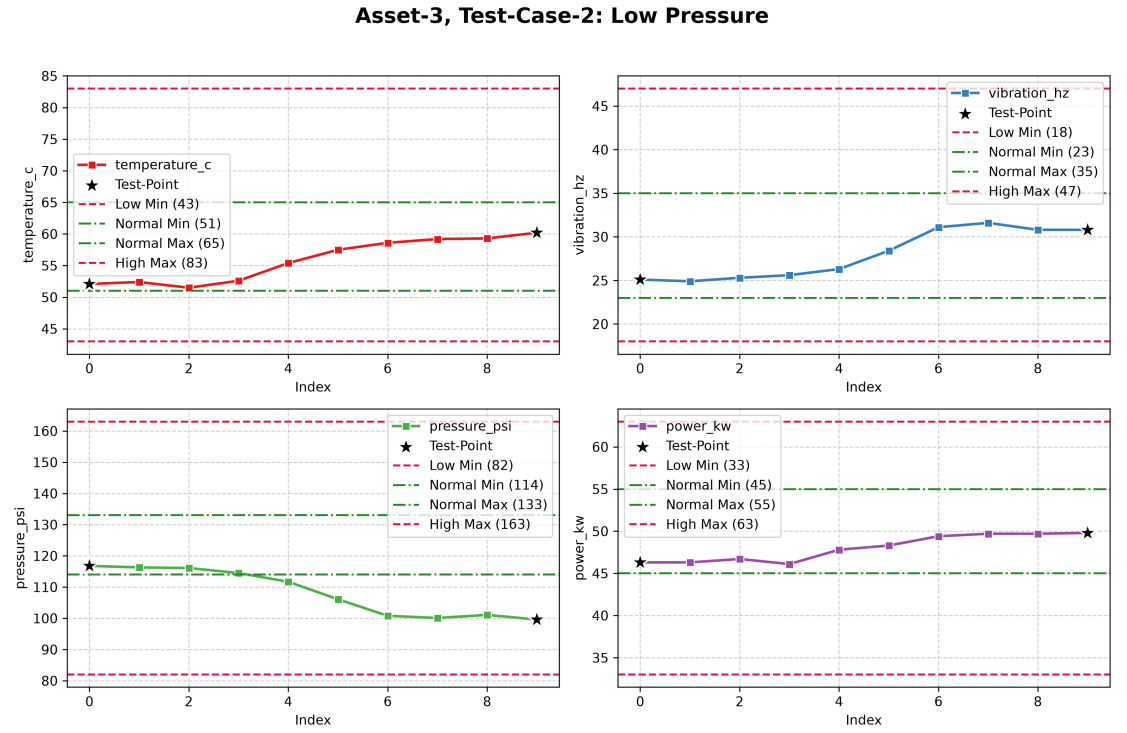
Sample test-case for anomaly `CLASS_0`.




```
max@m223:~/x/zc06-agentic-ai-sys
Updated 1 path from the index
max@m223:~/x/zc06-agentic-ai-sys$ ./use_ltm.sh 1 ; uv run agentic_system.py --asset 3 --test 2 --samples 2
[Check-dB] Telemetry
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_telemetry/telemetry.db (no action is taken)
[Check-dB] LTM
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_ltm/ltm.db (no action is taken)
[Check-dB] Checkpoint
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_checkpoint/checkpoint_db (no action is taken)
[Use LTM as Context] True
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[Running] Test-case-2 for asset-id:3 with 2 samples.
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[TELEMETRY] {'temperature_c': 52.1, 'vibration_hz': 25.0, 'pressure_psi': 115.9, 'power_kw': 46.4}
[RESULT] Not Anomaly
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[TELEMETRY] {'temperature_c': 59.9, 'vibration_hz': 31.6, 'pressure_psi': 99.9, 'power_kw': 49.6}
[RESULT] Anomaly: CLASS_1
Asset: Pump-3, CLASS_1-Anomaly detected at 2026-09-15 16:00:06
Telemetry: {'temperature_c': 59.9, 'vibration_hz': 31.6, 'pressure_psi': 99.9, 'power_kw': 49.6}
Severity Bands: {'temperature_c': 'Normal', 'vibration_hz': 'Normal', 'pressure_psi': 'Low', 'power_kw': 'Normal
'}
Failure: Normal Operation
Root Cause: All parameters are within normal operational ranges, indicating normal operation.
Confidence: 1.0
Suggested Action: No immediate action required. Continue monitoring.
Action Taken: set VFD control to increase pressure_psi
from 99.9-Low pressure to 123.5 (mean Normal value).
max@m223:~/x/zc06-agentic-ai-sys$
```

Figure 5

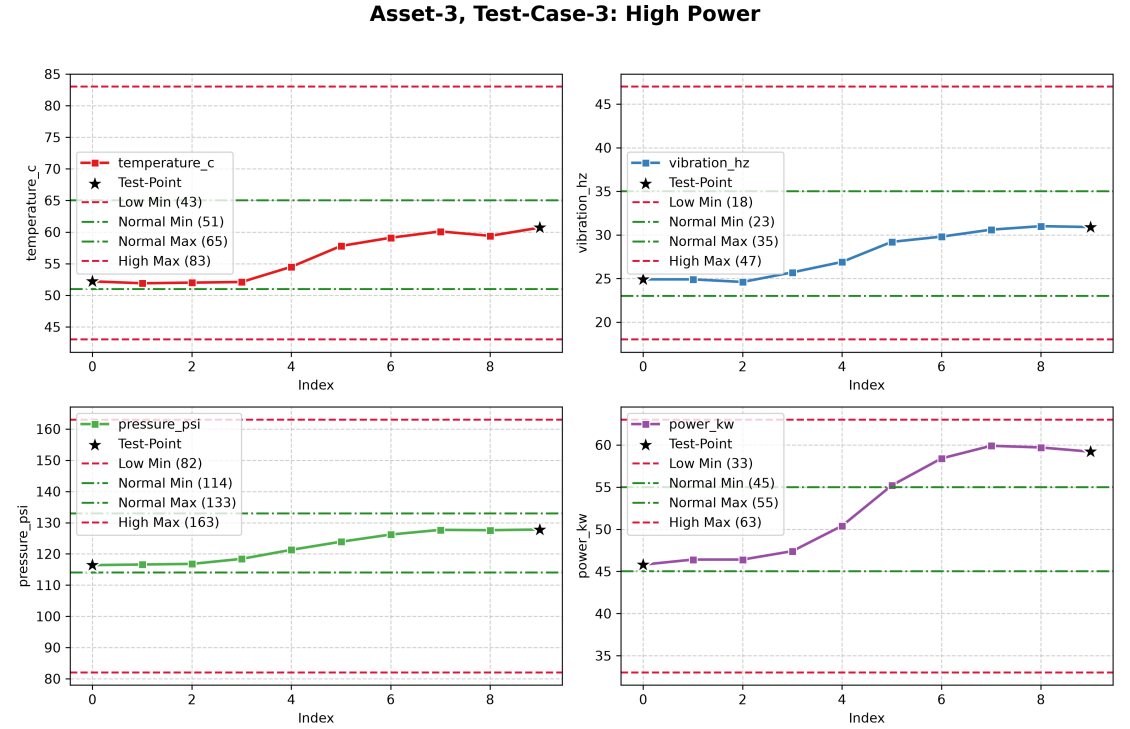
Sample test-case for anomaly `CLASS_1`.




```
max@m223:~/x/zc06-agentic-ai-sys
Updated 1 path from the index
max@m223:~/x/zc06-agentic-ai-sys$ ./use_ltm.sh 1 ; uv run agentic_system.py --asset 3 --test 2 --samples 2
[Check-dB] Telemetry
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_telemetry/telemetry.db (no action is taken)
[Check-dB] LTM
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_ltm/ltm.db (no action is taken)
[Check-dB] Checkpoint
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_checkpoint/checkpoint_db (no action is taken)
[Use LTM as Context] True
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[Running] Test-case-2 for asset-id:3 with 2 samples.
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[TELEMETRY] {'temperature_c': 52.1, 'vibration_hz': 25.0, 'pressure_psi': 115.9, 'power_kw': 46.4}
[RESULT] Not Anomaly
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[TELEMETRY] {'temperature_c': 59.9, 'vibration_hz': 31.6, 'pressure_psi': 99.9, 'power_kw': 49.6}
[RESULT] Anomaly: CLASS_1
Asset: Pump-3, CLASS_1-Anomaly detected at 2026-09-15 16:00:06
Telemetry: {'temperature_c': 59.9, 'vibration_hz': 31.6, 'pressure_psi': 99.9, 'power_kw': 49.6}
Severity Bands: {'temperature_c': 'Normal', 'vibration_hz': 'Normal', 'pressure_psi': 'Low', 'power_kw': 'Normal
'}
Failure: Normal Operation
Root Cause: All parameters are within normal operational ranges, indicating normal operation.
Confidence: 1.0
Suggested Action: No immediate action required. Continue monitoring.
Action Taken: set VFD control to increase pressure_psi
from 99.9-Low pressure to 123.5 (mean Normal value).
max@m223:~/x/zc06-agentic-ai-sys$
```

Figure 6

Sample test-case for anomaly `CLASS_2`.




```
max@m223:~/x/zc06-agentic-ai-sys
max@m223:~/x/zc06-agentic-ai-sys$ ./use_ltm.sh 0 ; uv run agentic_system.py --asset 3 --test 3 --samples 2
[Check-dB] Telemetry
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_telemetry/telemetry.db (no action is taken)
[Check-dB] LTM
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_ltm/ltm.db (no action is taken)
[Check-dB] Checkpoint
database already exist at :/home/max/x/zc06-agentic-ai-sys/out/db_checkpoint/checkpoint_db (no action is taken)
[Use LTM as Context] False
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[Running] Test-case-3 for asset-id:3 with 2 samples.
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[TELEMETRY] {'temperature_c': 52.1, 'vibration_hz': 25.0, 'pressure_psi': 115.9, 'power_kw': 46.4}
[RESULT] Not Anomaly
++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++++
[TELEMETRY] {'temperature_c': 59.9, 'vibration_hz': 31.6, 'pressure_psi': 127.9, 'power_kw': 59.6}
[RESULT] Anomaly: CLASS_1
Asset: Pump-3, CLASS_1-Anomaly detected at 2026-09-15 16:14:40
Telemetry: {'temperature_c': 59.9, 'vibration_hz': 31.6, 'pressure_psi': 127.9, 'power_kw': 59.6}
Severity Bands: {'temperature_c': 'Normal', 'vibration_hz': 'Normal', 'pressure_psi': 'Normal', 'power_kw': 'Hig
h'}
Failure: High Power Consumption
Root Cause: The power consumption is higher than normal, indicating potential overload or inefficiency in the pu
mp system.
Confidence: 1.0
Suggested Action: Monitor the pump closely, check for any abnormal operational conditions, and consider reducing
 load if necessary.
Action Taken: set VFD control to decrease power_kw
from 59.6-High power_kw to 50.0 (mean Normal value).
max@m223:~/x/zc06-agentic-ai-sys$
```

Figure 7

Sample test-case (Asset 3, Test 3) for anomaly `CLASS_1` *without LTM.*

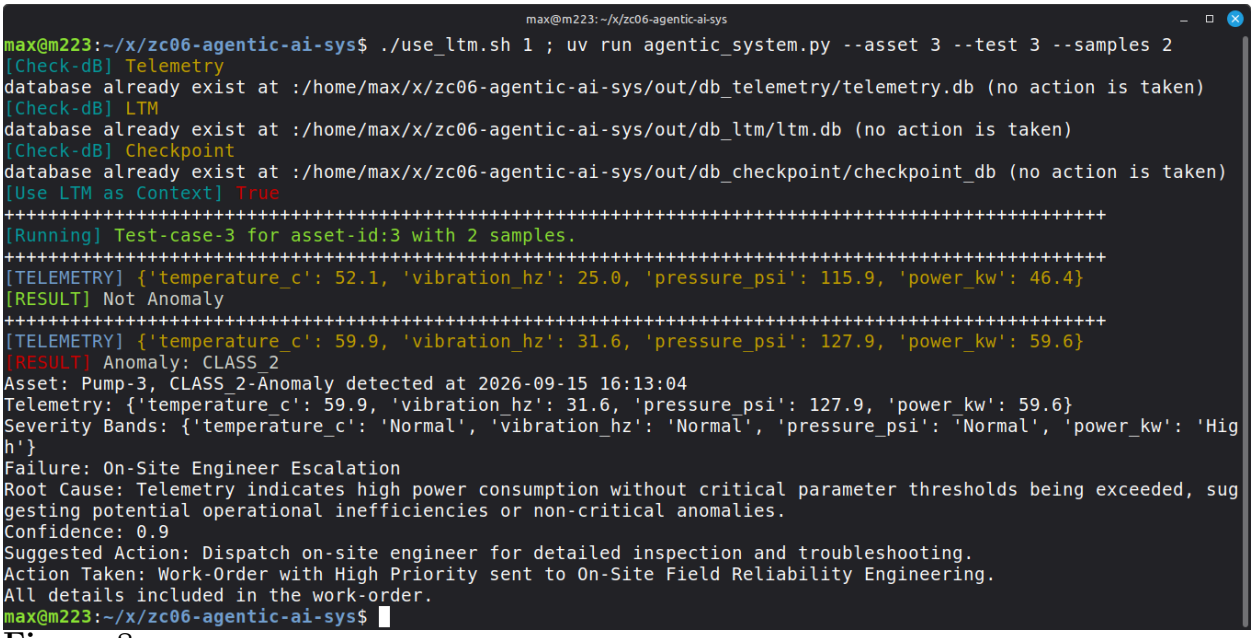


Figure 8

Sample test-case (Asset 3, Test 3) for anomaly ***CLASS_1*** *using LTM.*

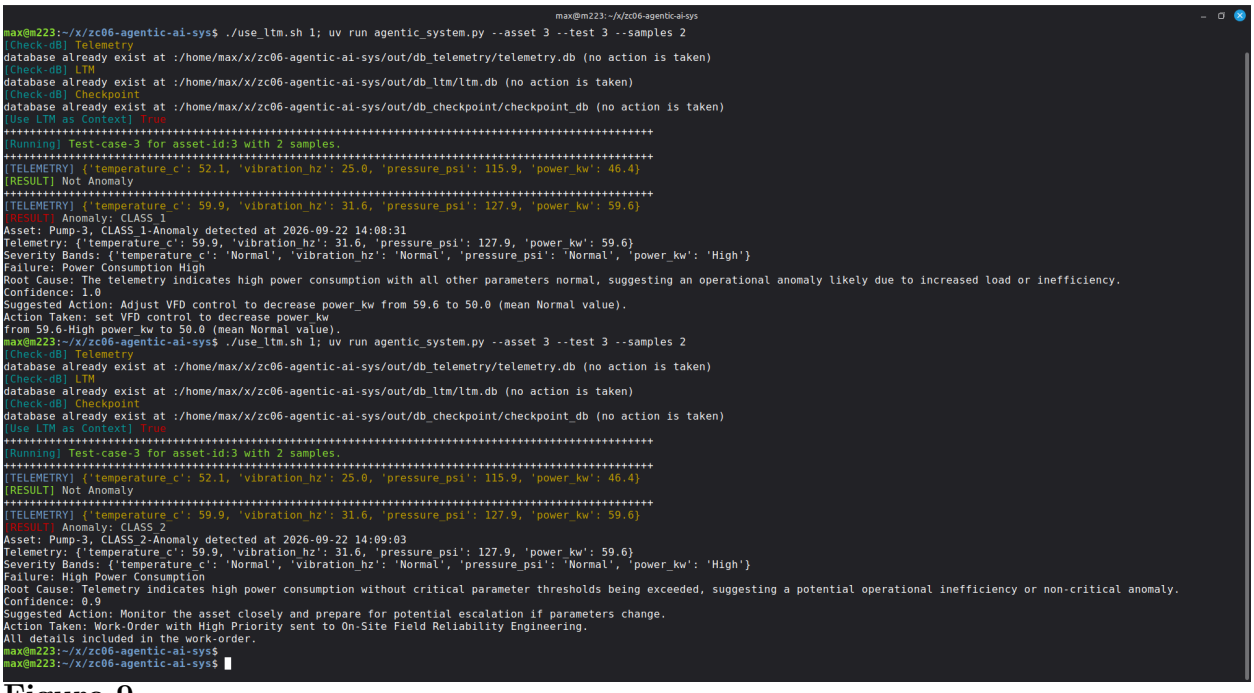


Figure 9

Sample test-case (Asset 3, Test 3) using LTM (Repeat after several days).

Ethical and Responsible Use Considerations

Automating control decisions and emergency shutdowns in industrial physical infrastructure introduces significant operational risks. Granting autonomous agents direct execution authority over `CLASS_0` emergency shutdowns poses a trade-off:

- **Safety vs. Financial Cost:** An automated emergency shutdown prevents catastrophic physical failure, protecting personnel and infrastructure. However, false positives cause expensive, unnecessary production halts.
- **Accountability and Delays:** Requiring a Human-in-the-Loop (HITL) for critical shutdowns mitigates false-positive costs but introduces latency that may lead to physical destruction during fast-moving failure modes.

Future Improvements

To address present system limitations, future development will target the following areas:

1. **Machine Learning Anomaly Detection:** Replace rule-based deviation bounds with trained machine learning models fitted on historical industrial telemetry.
2. **Generative Data Pipelines:** Integrate Generative AI models to synthesize diverse normal and rare failure-mode telemetry for robust benchmark testing.
3. **Expanded Actuator Integration:** Extend control capability beyond Variable Frequency Drives (VFDs) to include actuators like bypass and recirculation valves.
4. **Closed-Loop Simulation:** Implement continuous closed-loop feedback environments to dynamically evaluate asset recovery following agent-initiated control adjustments.
5. **Configurable Safety Guards:** Incorporate optional Human-in-the-Loop approval nodes for emergency shutdowns based on facility-specific risk profiles.

References

Carrascosa, I. P. (2026, June). *Context window management for long-running agents: Strategies and tradeoffs* [Accessed: 2026-09-24]. Machine Learning Mastery. https://machinelearningmastery.com/context-window-management-for-long-running-agents-strategies-and-tradeoffs/

Chugani, V. (2025, December). *Beyond short-term memory: The 3 types of long-term memory AI agents need* [Accessed: 2026-09-22]. Machine Learning Mastery. https://machinelearningmastery.com/beyond-short-term-memory-the-3-types-of-long-term-memory-ai-agents-need/

Gadesha, V. (2026). What is few shot prompting? [Accessed: 2026-09-22]. *IBM*. https://www.ibm.com/think/topics/few-shot-prompting

Hughes, L., Dwivedi, Y. K., Malik, T., Shawosh, M., Albashrawi, M. A., Jeon, I., Dutot, V., Appanderanda, M., Crick, T., De', R., Fenwick, M., Gunaratnege, S. M., Jurcys, P., Kar, A. K., Kshetri, N., Li, K., Mutasa, S., Samothrakis, S., Wade, M., & Walton, P. (2025). Ai agents and agentic systems: A multi-expert analysis. *Journal of Computer Information Systems*, *65*(4), 489–517. https://doi.org/10.1080/08874417.2025.2483832

LangChain Inc. (2026). Langgraph overview [Accessed: 2026-09-15]. https://docs.langchain.com/oss/python/langgraph/overview

Plaat, A., Van Duijn, M., Van Stein, N., Preuss, M., Van der Putten, P., & Batenburg, K. J. (2025). Agentic large language models, a survey. *Journal of Artificial Intelligence Research*, *84*. https://doi.org/10.1613/jair.1.18675

Stryker, C. (2026). What is AI agent memory? [Accessed: 2026-09-22]. *IBM*. https://www.ibm.com/think/topics/ai-agent-memory

Wikipedia contributors. (2026a). Centrifugal pump — Wikipedia, the free encyclopedia [Online; accessed 16-September-2026]. https://en.wikipedia.org/w/index.php?title=Centrifugal_pump&oldid=1373998212

Wikipedia contributors. (2026b). Large language model — Wikipedia, the free encyclopedia [Online; accessed 15-September-2026]. https://en.wikipedia.org/w/index.php?title=Large_language_model&oldid=1374737380

Wikipedia contributors. (2026c). Retrieval-Augmented Generation (RAG) — Wikipedia, the free encyclopedia [Online; accessed 22-September-2026]. https://en.wikipedia.org/w/index.php?title=Retrieval-augmented_generation&oldid=1375082076

Wikipedia contributors. (2026d). Variable-Frequency Drive — Wikipedia, the free encyclopedia [Online; accessed 24-September-2026]. https://en.wikipedia.org/w/index.php?title=Variable-frequency_drive&oldid=1367459418

Yao, S., Zhao, J., Yu, D., Du, N., Shafran, I., Narasimhan, K., & Cao, Y. (2023). ReAct: Synergizing Reasoning and Acting in Language Models. https://arxiv.org/abs/2210.03629

Appendix

Appendix A: Asset and Test-case Info

Table A1 shows severity bands thresholds for 3 assets used in this project, these thresholds will determin the normal and anomaly region as shown in Fig. 2 .

Table A1

Asset Severity Band Thresholds.

asset_index	parameter	low_min	normal_min	normal_max	high_max
1	temperature_c	40	48	62	80
1	vibration_hz	15	20	32	44
1	pressure_psi	80	111	130	160
1	power_kw	30	42	52	60
2	temperature_c	41	49	63	81
2	vibration_hz	16	21	33	45
2	pressure_psi	81	112	131	161
2	power_kw	31	43	53	61
3	temperature_c	43	51	65	83
3	vibration_hz	18	23	35	47
3	pressure_psi	82	114	133	163
3	power_kw	33	45	55	63

Table A2 shows the test-case data (3 cases per asset, total 9 scenarios). Each test case defined by start and end values for all 4 telemetry. we use starting point in normal range, and the end point somewhere in anomaly region. For simulation and test these two points are enough, but we can create a trend between these two points and send all of them to the model for evaluation, For example using asset-3 with test case-1, and 5 samples shown in Fig. A1, where we can see:

- All telemetry except pressure are within normal range.
- First 2 samples of pressure are in normal range.
- 3rd sample is high-pressure which is CLASS_1 anomaly as detected by Agent.
- 4th and 5th samples are critically high which is CLASS_0 as detected by Agent.

Table A2

Test Cases Data

asset_index	parameter	start1	end1	start2	end2	start3	end3
1	temperature_c	50	70	50	79	50	90
1	vibration_hz	22	29	22	29	22	29
1	pressure_psi	114	125	114	135	114	125
1	power_kw	44	48	44	48	44	48
2	temperature_c	51	60	51	60	51	60
2	vibration_hz	23	35	23	40	23	50
2	pressure_psi	116	129	116	140	116	129
2	power_kw	45	49	45	49	45	49
3	temperature_c	52	60	52	60	52	60
3	vibration_hz	25	31	25	31	25	31
3	pressure_psi	117	180	117	100	117	128
3	power_kw	46	50	46	50	46	60

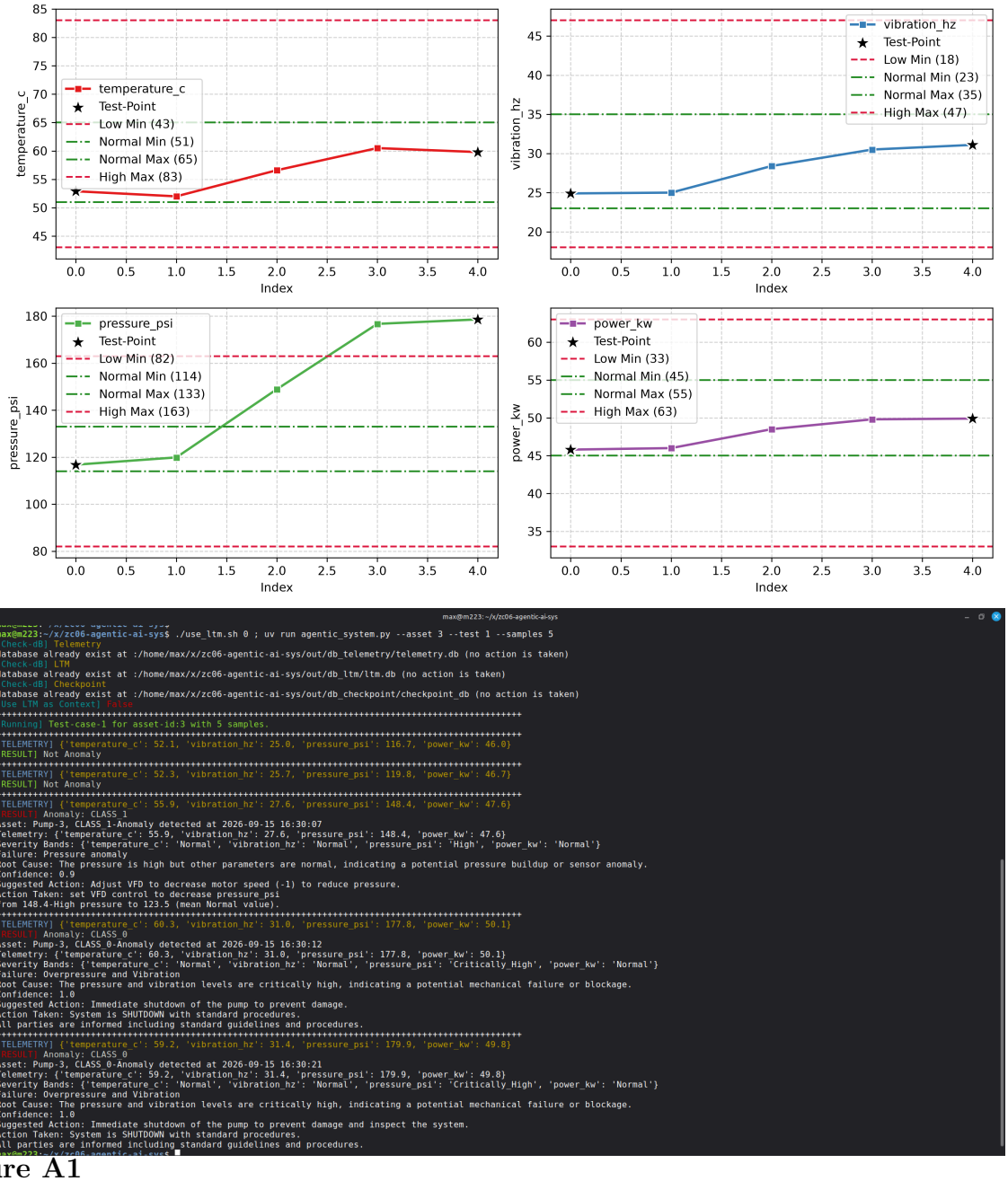


Figure A1

Sample test-case with 5 samples.